

corner, and the cells and fine picture details utilise a number of cells to-

wards the bottom right-hand corner.

The grey-scale pattern is shown

of the DCT coefficients is not uniform.

An average DCT matrix has most of its coefficients—and therefore energy—concentrated at and around the top left-hand corner. The bottom right-hand quadrant has very few coefficients of any substantial value. Bit rate reduction may thus be achieved by not transmitting the zero and near-zero coefficients.

Further bit reduction

may be introduced by weighted quantising and special coding techniques of the remaining coefficients.

Quantising the DCT block. After a block has been transformed, the DCT coefficients are quantised (rounded up or down) to a smaller set of possible values to produce a simplified set of coefficients.

The DCT block (shown in Fig. 10(a)) may be reduced to very few coefficients (Fig. 10(b)) if a threshold of 1.0 is applied. After this, a non-linear or weighted quantisation is applied. The video samples are given a linear quantisation but the DCT coefficients receive a non-linear quantisation. A

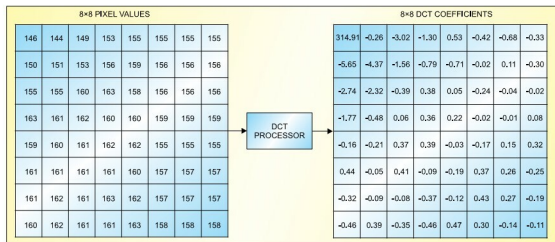


Fig. 7: 8x8-pixel block transformed into 8x8 DCT coefficient block

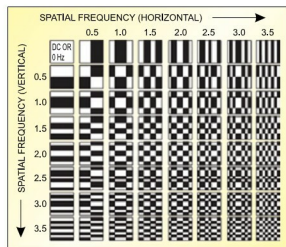


Fig. 8: DCT wave table illustrating the spatial frequencies represented by each DCT cell

in Fig. 9(a). The corresponding sample values are shown in Fig. 9(b) and DCT coefficients in Fig. 9(c). DCT does not directly reduce the number of bits required to represent the 8x8-pixel block. Sixty-four pixel sample values are replaced with 64 DCT coefficient values.

The reduction in the number of bits follows from the fact that for a typical block of a natural image, the distribution

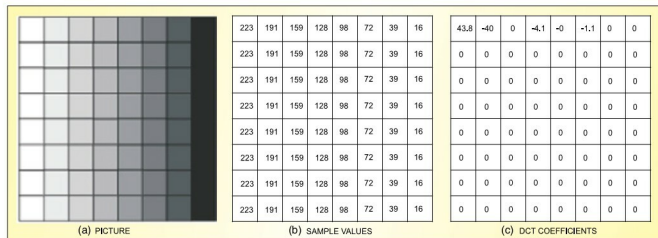


Fig. 9: DCT coefficients of a grey-scale 8x8 block